Labs 7, 10, and 11

Combined Docker demonstration submission

One Compose project

The single docker-compose.yml starts the Lab 7 Flask app, the Lab 10 replica set, and the Lab 11 two-shard cluster with its config server and mongos router. The Lab 11 runner uses the same project and image.

Lab	Included work
7	CTI report research desk, dataset, search and classification app, and working-set/cache exercises.
10	Bank transfers with transaction rollback and retries; concurrent inventory reservations and throughput comparison.
11	Shard key decision matrix, MongoDB 7 two-shard hotspot comparison, explain plans, and resharding research.

Start and demonstrate

```
docker compose up --build -d
docker compose run --rm app python scripts/lab10_transactions.py all
docker compose --profile lab11 run --rm lab11
```

Published Docker image

Docker Hub: krish1226/lab10 | **Tag:** lab11

```
docker pull krish1226/lab10:lab11
```

The Lab 11 runner produces shard-by-shard counts and query plans in docs/lab11-results.json. Detailed instructions and design analysis are in docs/lab10.md and docs/lab11.md.